# SMS Spam Classification Database Creation

An AI-driven Approach to Combat Unsolicited Messages

# Flow of Presentation

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# Problem Statement

The rapid growth of mobile communication has led to a surge in SMS messages, many of which are unsolicited or spam. Spam messages not only waste user time but also pose risks such as phishing, scams, and malware links. The problem is to design a system that can automatically classify SMS messages into spam or non-spam categories by using computational methods, reducing the need for manual intervention and improving user safety.

# Evolution of Spam Classification

**A. Rule-based Approaches**

Early SMS filtering methods used keyword spotting and rule-based heuristics.
*Limitation*: Easily bypassed by variations in wording and required manual updates.

**B. Traditional Machine Learning**

Algorithms like Naïve Bayes, SVM, and Decision Trees have been widely applied.
They rely on features such as bag-of-words, TF-IDF, and n-grams.
Works well with labeled datasets but performance depends heavily on feature engineering.

**C. Deep Learning Methods**

CNNs and RNNs capture sequential and contextual patterns in messages.
Better at handling variations in spam text compared to traditional models.
However, they require large amounts of labeled data and high computational resources.

**D. Transfer Learning & Pre-trained Models**

Recent approaches leverage models like BERT and other transformers.
They achieve state-of-the-art performance by understanding semantic context.
The drawback is that training and fine-tuning are computationally expensive.

# Evolution of Spam Classification

1990s

## Simple Keyword Rules

Early spam detection used basic keyword and phrase filters, blocking emails containing words like "Viagra" or "free money." While straightforward, these rule-based systems were easily outsmarted by...

Early 2000s

## Probabilistic Spam Detection

Bayesian filters introduced a probabilistic method, learning from users' mail to identify spam based on word frequencies. This adaptive technique significantly improved accuracy by calculating the ...

Mid 2000s

## Advanced ML Techniques

The adoption of machine learning algorithms like Support Vector Machines (SVMs) and decision trees enabled more sophisticated analysis. These models identified complex patterns, header anomalies,etc.

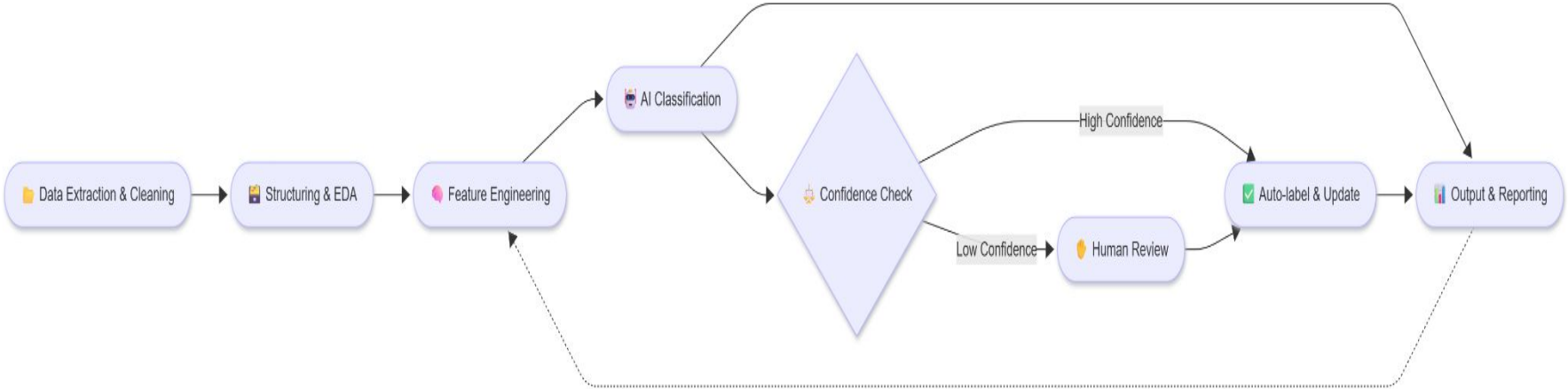
# Expected Outcome/Objective

- High-quality, labeled SMS dataset
- Spam/Ham + subcategories (phishing, bank, promotional)
- Cleaned, balanced, anonymized
- ML-ready format (CSV)
- Data dictionary & documentation
- Exploratory Data Analysis
- Baseline model (optional)
- Reliable, ethical, versatile

# Task to Accomplish

- Extract SMS messages from backups
- Process messages into structured datasets
- Apply automated classification to identify spam and scam
- Reduce manual effort and ensure consistency
- Create a scalable solution for large SMS datasets

# Workflow – Diagram – 2



# Parsing SMS Data from XML Backup

- Extract raw SMS data from structured XML files generated by mobile backup apps
- Identify and retrieve key fields: sender info, timestamp, message body, message type (Inbox, Sent, Draft, etc.)
- Clean extracted data by removing duplicates, special characters, and inconsistencies
- Convert timestamps to standard date-time format
- Organize data into structured format (e.g., Pandas DataFrame)
- Prepare dataset for further analysis, feature extraction, and automated classification

[6]:
```
df
```

[6]:

| | date | time | sender | message_type | message_body | spam |
|---|---|---|---|---|---|---|
| **0** | 2025-07-29 | 13:36:08 | BV-INPOST-G | Inbox | Article PT071224430IN delivered on 29/07/2025 ... | 0 |
| **1** | 2025-07-29 | 19:04:14 | AD-SBIUPI-S | Inbox | Dear SBI User, your A/c X7447-credited by Rs.4... | 0 |
| **2** | 2025-07-30 | 11:28:28 | VM-SBIUPI-S | Inbox | Dear UPI user A/C X7447 debited by 48000.0 on ... | 0 |
| **3** | 2025-07-30 | 11:31:11 | AX-SBIINB-S | Inbox | Dear Customer, SBCollect Txn DUO4034076 dtd 30... | 0 |
| **4** | 2025-07-30 | 15:41:49 | VA-CANBNK-S | Inbox | An amount of INR 16,300.00 has been DEBITED to... | 0 |
| **...** | ... | ... | ... | ... | ... | ... |
| **119** | 2025-09-01 | 16:43:50 | AX-CANBNK-S | Inbox | An amount of INR 3,368.00 has been DEBITED to ... | 0 |
| **120** | 2025-09-01 | 16:44:13 | AX-CANBNK-S | Inbox | Dear Customer,An advance amount of Rs.INR 3,36... | 0 |
| **121** | 2025-09-01 | 18:38:58 | AE-AIRTEL-S | Inbox | रिमाइंडर: आपके पास 1GB डाटा लोन का भुगतान करना... | 0 |
| **122** | 2025-09-01 | 20:08:47 | AE-AIRTEL-S | Inbox | हम हर दिन बेहतर हो रहे हैं! BAKTAURI PURWA में... | 0 |
| **123** | 2025-09-01 | 20:39:41 | AE-AIRTEL-S | Inbox | हम हर दिन बेहतर हो रहे हैं! WAKTORI में एक नया... | 0 |

124 rows × 6 columns

# SMS Classification

- Load cleaned SMS dataset (CSV format) with message body and metadata
- Categorize messages into spam subtypes (Phishing, Smishing, Promotional, Loan Scam, etc.) and no spam
- Preprocess text data to normalize and prepare messages for classification
- Use API-based ensemble models to classify SMS with confidence scoring
- Implement retry mechanisms to handle API rate limits gracefully
- Save the annotated dataset with predicted labels and confidence scores for further analysis and model evaluation

*Note: This approach is necessary due to limited available Indian SMS spam datasets.*

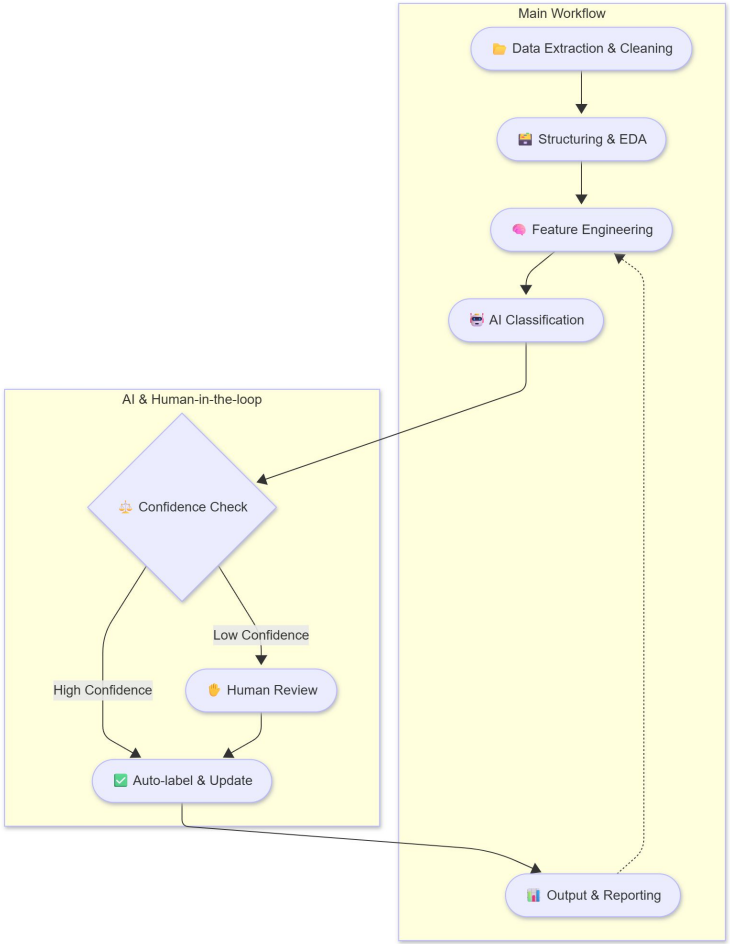
# Models and Approach

Utilized multiple pre-trained language models via Ollama API:

- deepseek-v3.1671b-cloud
- qwen3–coder480b-cloud
- gpt-oss20b-cloud
- gpt-oss120b-cloud

- Classify SMS into key categories (Phishing, Smishing, Promotional, Loan/Financial Scam, Job Scam, Crypto/Investment Scam, No Spam)
- Generate confidence scores for each category
- Aggregate predictions using majority voting and average confidence
- Flag low-confidence messages for human review to refine dataset quality

# Workflow Diagram



# Challenged Faced

- API rate limits restricted SMS classifications per minute, slowing processing.
- Variations in XML SMS backup formats caused parsing errors.
- Missing or inconsistent timestamps required extra error handling.
- Large datasets increased processing time, necessitating automation.
- Offline models (e.g., LLaMA) were inefficient at scale.
- Ambiguous SMS content complicated classification accuracy.
- Managing multiple spam categories needed careful label definition.
- Ensuring data privacy and anonymization added preprocessing steps.

# Application

- Mobile spam filtering systems to protect users from scams, phishing, and fraudulent messages.
- Telecom providers analyzing SMS data to detect and prevent fraudulent activity.
- Enterprise communication monitoring to identify and block phishing or scam attempts.
- Research projects using SMS datasets for natural language processing, model training, and analysis.
- Fraud detection in banking and financial services through automated SMS monitoring.
- Marketing analytics to study user engagement and trends while filtering unwanted messages.
- Example: Airtel AI uses machine learning to automatically detect and filter spam messages for its users.

# Limitations

- Classification depends on external models with rate limits and usage restrictions.
- Ambiguous or context-dependent SMS content can cause misclassifications.
- Variations in XML SMS backup format limit parser adaptability.
- Current system is batch-only; no real-time filtering supported.
- Offline models (e.g., LLaMA) inefficient for large datasets.
- Large datasets increase processing time and resource demands.
- Privacy and anonymization add preprocessing complexity.
- Limited generalization to unseen SMS formats or languages.
- Dataset completeness affected as telecom spam filters block many messages before delivery.

# Why Automated Classification is Needed

- The rapid increase in SMS traffic overwhelms manual spam filtering.
- Automation quickly and accurately distinguishes legitimate vs. malicious messages.
- Protects users from scams, phishing, and unwanted promotions.
- Reduces human effort and ensures consistent labeling.
- Enables scalable, real-time spam detection.
- Machine learning adapts to evolving spam tactics for improved security and user experience.
- Efficiently processes massive datasets with high accuracy and speed.

# Conclusion

This project demonstrates how raw SMS data is effectively transformed into structured datasets and automatically classified into spam and scam categories. By integrating parsing, cleaning, and classification, the workflow handles large volumes of messages efficiently without manual labeling. Automation proved essential to overcome challenges of data inconsistency, high manual effort, and time constraints. Overall, this approach is practical and scalable for large-scale SMS analysis and spam detection.

# References

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# THANK YOU